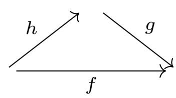
Chapter 10

Topos

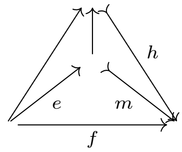
10.1 Factorization and Images

Definition 10.1.1. A morphism f *factors through* a morphism g if there is a morphims h such that $f = g \circ h$.



Definition 10.1.2 (Image)**.** A monic m is the *image* of the arrow f if the following conditions are satisfied.

- f factors through m, i.e., $f = m \circ e$ for some e.
- Whenever f factors through a monic h, so does m.



Remark 10.1.3. Definition 10.1.2 says that the image m is a smallest subobject (of the codomain of f) through which f can factor.

Theorem 10.1.4 (Epi-mono factorization)**.** If a category $\mathcal{C}$ has a cokernel pair for any morphism, then every arrow f in $\mathcal{C}$ has an image m and factors as $f = m \circ e$ with e epic.

Remark 10.1.5. By Theorem 10.4.1, the result of Theorem 10.1.4 holds for any topos.

Proof. MacLane and Moerdijk (1992, p.185)   to be written

Given $f : A \longrightarrow B$, let x, y be the cokernel pair of f and $m : M \longrightarrow B$ an equalizer of x, y. Then, there must be a unique arrow from A to M since $x \circ f = y \circ f$.

$$\begin{tikzcd}
A \arrow[rr, "f"] \arrow[dr, dotted, "!"'] & & B \arrow[r, shift left, "x"] \arrow[r, shift right, "y"'] & {} \\
& M \arrow[ur, "m"'] & &
\end{tikzcd}$$

By Theorem 2.4.2, m is monic.

Now, take other factorization $f = h \circ g$ with h monic.

$$\begin{tikzcd}
A \arrow[rr, "f"] \arrow[dr, dotted, "!"] \arrow[ddr, "g"'] & & B \arrow[r, shift left, "x"] \arrow[r, shift right, "y"'] & {} \\
& M \arrow[ur, tail, "m"] & & \\
& N \arrow[uur, tail, "h"'] & &
\end{tikzcd}$$

Theorem 2.5.4 ∎

10.2 Sub-objects

Definition 10.2.1 (Membership)**.** A morphism $x : T \longrightarrow A$ is a *member* of a monic $i : S \rightarrowtail A$ if there is some $h : T \longrightarrow S$ such that $s = i \circ h$.

$$\begin{tikzcd}
T \arrow[r, "h"] \arrow[dr, "x"'] & S \arrow[d, tail, "i"] \\
& A
\end{tikzcd}$$

Definition 10.2.2 (Inclusion)**.** Given two monics $i : I \rightarrowtail A$ and $j : J \rightarrowtail A$, i is *included* in j (notation $i \subseteq j$) if there is an arrow s (called an *inclusion arrow*) such that $i = j \circ s$.

$$\begin{tikzcd} I \arrow[rr, "s"] \arrow[dr, tail, "i"'] & & J \arrow[dl, tail, "j"] \\ & A & \end{tikzcd}$$

Theorem 10.2.3. For any two monics $i : I \rightarrowtail A$ and $j : J \rightarrowtail A$, if there exists an inclusion arrow, it is unique and monic.

Proof. If $i = j \circ s = j \circ s'$, then $s = s'$ since j is monic.

For any inclusion arrow s, if $s \circ h = s \circ k$, then $j \circ s \circ h = j \circ s \circ k$. By the assumption $i = j \circ s$, $i \circ h = i \circ k$. Therefore $h = k$ since i is monic. $\square$

Definition 10.2.4 (Intersection)**.** If two monics $i : I \rightarrowtail A$ and $j : J \rightarrowtail A$ has a pullback $I \cap J$, an *intersection* of i and j (notation $i \cap j$) is composites along either side from $I \cap J$ to A.

$$\begin{tikzcd} I \cap J \arrow[r] \arrow[d] \arrow[dr, "i \cap j"] & J \arrow[d, tail, "j"] \\ I \arrow[r, tail, "i"'] & A \end{tikzcd}$$

Now we are ready to define the union of two sub-objects.

Definition 10.2.5 (Union)**.** If two monics $i : I \rightarrowtail A$ and $j : J \rightarrowtail A$ has a coproduct $I + J$, a *union* of i and j (notation $i \cup j$) is a image of the unique arrow from a selected coproduct $I + J$ to A.

$$\begin{tikzcd} I + J \arrow[rr] \arrow[dd] \arrow[dr, "e"] & & J \arrow[dd, tail, "j"] \\ & {} \arrow[dr, tail, "i \cup j"] & \\ I \arrow[rr, tail, "i"'] & & A \end{tikzcd}$$

Definition 10.2.6 (Complement)**.** For a monic $i : I \rightarrowtail A$, a *complement for i* is a monic $j : J \rightarrowtail A$ such that $I \cap J \cong \emptyset$ and $I \cup J \cong 1$.

Theorem 10.2.7. For any two monics $i : I \rightarrowtail A$ and $j : J \rightarrowtail A$, $i \subseteq j$ iff $i = i \cap j$.

Proof. ($\leftarrow$) Obvious.

(→) Since $i \subseteq j$, there is unique s such that the following diagram commutes.

$$\begin{tikzcd} I \arrow[r, "s"] \arrow[d, "id"'] & J \arrow[d, tail, "j"] \\ I \arrow[r, tail, "i"'] & A \end{tikzcd}$$

If there is any object T and morphisms h and k such that the following diagram commutes, $h = k$ since s is monic according to Theorem 10.2.3.

$$\begin{tikzcd} T \arrow[d, shift right, "h"'] \arrow[d, shift left, "k"] \arrow[dr] & \\ I \arrow[r, "s"'] & J \end{tikzcd}$$

Therefore I is a pullback of I and J, namely $I \cap J$.

□

Theorem 10.2.8. Subobjets in a Topos is a Heyting algebra.

10.3 Sub-object Classifier

Definition 10.3.1 (Sub-object classifier)**.** A *sub-object classifier* is an object Ω and a global element $t : 1 \longrightarrow \Omega$ such that for any monic $s : S \rightarrowtail A$ there is a unique arrow $\chi_s : A \longrightarrow \Omega$ that makes the following a pullback:

$$\begin{tikzcd} S \arrow[r] \arrow[d, tail, "s"'] & 1 \arrow[d, "t"] \\ A \arrow[r] & \Omega \end{tikzcd}$$

Any arrow $A \longrightarrow \Omega$ corresponds to equivalence classes of sub-objects of A. For any arrow $w : A \longrightarrow \Omega$, we write $w^* : W \rightarrowtail A$ for the selected monic that w classifies.

Definition 10.3.2 (Topos)**.** A *topos* is a cartesian closed category with a sub-object classifier.

We will often use the abbreviation t_T for the composite $t\circ !_T$, and even omit the subscript T when it is clear from the context.

Theorem 10.3.3. For any $s : A \longrightarrow T$, $t_T \circ s = t_A$.

Proof.

$$t_T \circ s = t\circ !_T \circ s = t\circ !_A = t_A$$

□

Theorem 10.3.4. t_T classifies id_T.

Proof. The following diagram commutes.

$$\begin{tikzcd} T \arrow[r, "!_T"] \arrow[d, "id_T"'] & 1 \arrow[d, "t"] \\ T \arrow[r, "t_T"'] & \Omega \end{tikzcd}$$

By Theorem 10.3.4, $t_T \circ f = t_A$ for any morphism $f : A \longrightarrow T$.

$$\begin{tikzcd} A \arrow[dr] \arrow[drr, bend left, "!_A"] \arrow[ddr, bend right, "f"'] & & \\ & T \arrow[r, "!_T"] \arrow[d] & 1 \arrow[d, "t"] \\ & T \arrow[r, "t_T"'] & \Omega \end{tikzcd}$$

The arrow from A to T is unique since id_T is monic, that is, f itself. Thus the upper T is a pullback. □

10.4 Topos and Colimits

Theorem 10.4.1. A topos has all finite colimits.

Proof. to be written MacLane and Moerdijk (1992, p.183) Monad □

10.5 Alternative Axioms for Topos

$Hom_{\epsilon}(A \times B, C) \cong Sub()$

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